

Business Problem Statement

A retail organization wants to analyze customer purchasing behavior to identify opportunities for increasing revenue, improving customer retention, and making better marketing decisions.

Customers show different purchasing patterns across demographics, product categories, seasons, discount usage, and subscription behavior. The business requires a data-driven analysis to understand the key factors influencing customer decisions and repeat purchases.

The objective of this project is to analyze customer shopping data and uncover insights related to customer segments, product performance, spending patterns, and loyalty behavior. These insights will help the business optimize marketing strategies, improve customer engagement, and support strategic decision-making.

Key Business Question

How can customer transaction data be analyzed to identify shopping trends, improve customer retention, and optimize marketing and product strategies?

Project Deliverables

1. Data Preparation & Transformation (Excel + Power Query)

Clean and transform the raw customer dataset into an analysis-ready format by handling data quality issues, standardizing information, and creating required analytical features.

2. Data Analysis (MySQL)

Design and execute SQL analysis to identify customer behavior patterns, revenue trends, product performance, discount effectiveness, subscription impact, and customer loyalty insights.

3. Dashboard Development (Power BI)

Develop an interactive dashboard to visualize key performance indicators, customer trends, sales patterns, and business insights for stakeholders.

4. Business Report & Recommendations

Prepare a report summarizing analytical findings and provide actionable recommendations to improve marketing decisions, customer retention, and overall business performance.

5. Project Repository

Create a structured repository containing datasets, transformation files, SQL scripts, dashboard files, and complete project documentation.